Experiment No. 2

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Aim

To analyze and implement contrast enhancement techniques using histogram analysis and histogram equalization in Python and OpenCV for improving image quality and enhancing visual information in low-contrast images.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the low-contrast image
img = cv2.imread("low_contrast_image.jpg")

# Convert BGR to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

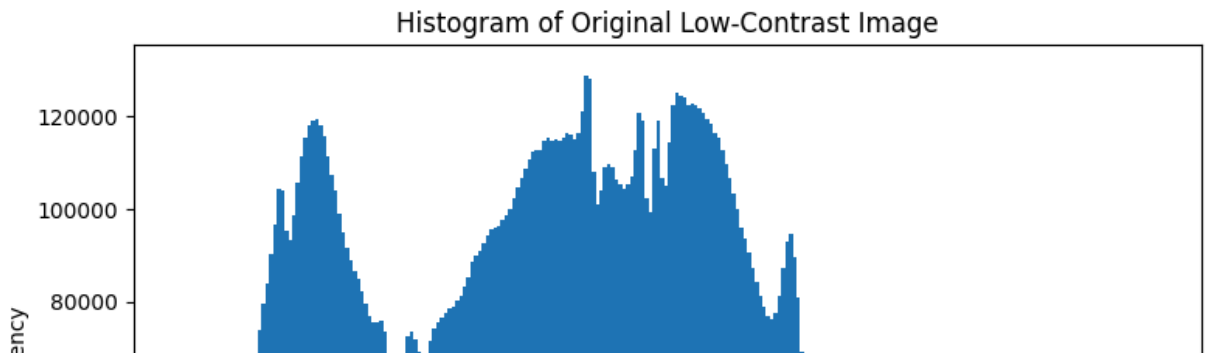
# Display original grayscale image
plt.figure(figsize=(8, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Low-Contrast Image")
plt.axis("off")
plt.show()
```



```
plt.figure(figsize=(8, 5))

plt.hist(
    gray.ravel(),
    bins=256,
    range=(0, 256)
)

plt.title("Histogram of Original Low-Contrast Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```




```
# Find minimum and maximum pixel intensity
min_val = np.min(gray)
max_val = np.max(gray)

print("Minimum pixel intensity:", min_val)
print("Maximum pixel intensity:", max_val)
```

```
# Apply contrast stretching
stretched = cv2.normalize(
    gray,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

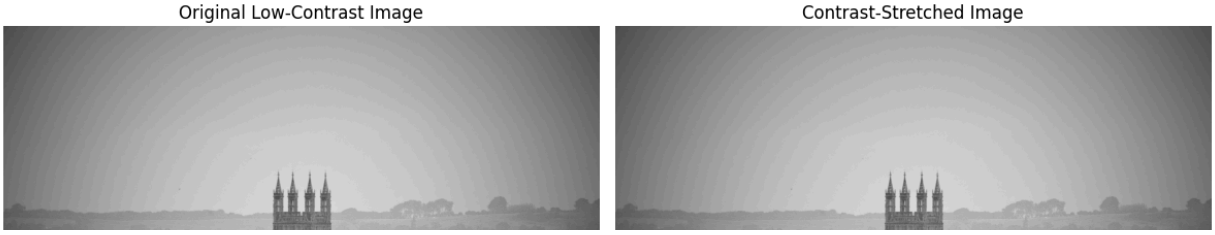
# Display original and contrast-stretched images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Low-Contrast Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(stretched, cmap="gray")
plt.title("Contrast-Stretched Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

```
Minimum pixel intensity: 0
Maximum pixel intensity: 198
```

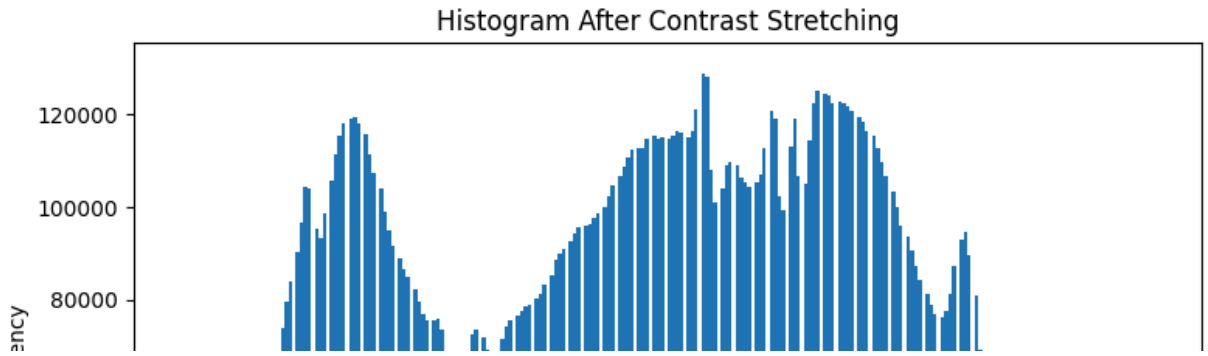



```
plt.figure(figsize=(8, 5))

plt.hist(
    stretched.ravel(),
```

```
    bins=256,
    range=(0, 256)
)

plt.title("Histogram After Contrast Stretching")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```




```
# Apply standard Histogram Equalization
equalized = cv2.equalizeHist(gray)

# Display original, contrast-stretched, and equalized images
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(equalized, cmap="gray")
```

```
plt.title("Histogram Equalization")
plt.axis("off")

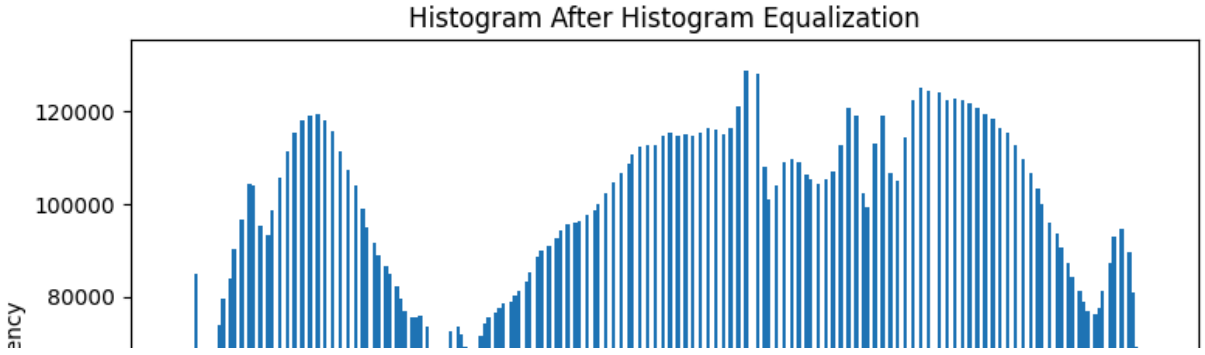
plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(8, 5))

plt.hist(
    equalized.ravel(),
    bins=256,
    range=(0, 256)
)

plt.title("Histogram After Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



```
# Create CLAHE object
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

# Apply CLAHE
clahe_image = clahe.apply(gray)

# Display comparison
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
```



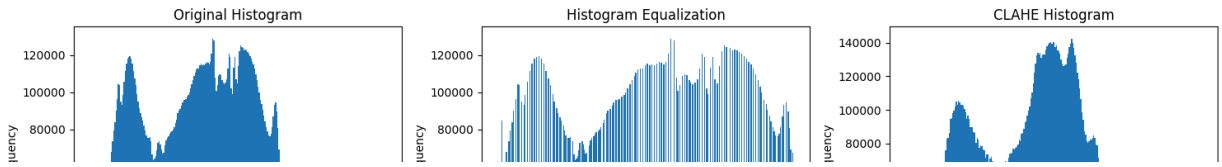
```
plt.figure(figsize=(15, 4))

plt.subplot(1, 3, 1)
plt.hist(gray.ravel(), bins=256, range=(0, 256))
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 3, 2)
plt.hist(equalized.ravel(), bins=256, range=(0, 256))
plt.title("Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 3, 3)
plt.hist(clahe_image.ravel(), bins=256, range=(0, 256))
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Visual Analysis

The original image has low contrast, which makes some details difficult to distinguish. After contrast stretching, the intensity range is expanded and the overall contrast of the image is improved.

Histogram Equalization further enhances the contrast by redistributing the pixel intensity values across the available range. This makes several image details more visible compared with the original image.

CLAHE improves local contrast by processing smaller regions of the image. In areas with uneven illumination, it can reveal local details while avoiding excessive enhancement in

some regions.

Overall, all three enhancement techniques improve the visibility of the original low-contrast image, but their effects differ depending on the intensity distribution and illumination conditions.

Comparison of Histogram Equalization and CLAHE

Histogram Equalization applies contrast enhancement globally across the entire image. It is useful when the overall intensity distribution needs improvement.

CLAHE works on local regions of the image and limits excessive contrast enhancement. This makes it useful for images where illumination varies across different areas.

For the given low-contrast image, Histogram Equalization provides overall contrast enhancement, while CLAHE provides more localized enhancement and can preserve local image details more effectively.

The choice between the two techniques depends on the type of image, its illumination conditions, and the amount of local detail that needs to be preserved.

QUESTIONS

Q1. What is image contrast, and why is it important in image processing?

Answer:

Image contrast refers to the difference in intensity between the darker and brighter regions of an image. High-contrast images have clearly distinguishable dark and bright areas, while low-contrast images may appear dull and unclear.

Contrast is important because it improves the visibility of important image details. Better contrast can make features easier to identify and can improve the quality of further image processing and computer vision tasks.

Q2. Explain the concept of an image histogram. What information does it provide?

Answer:

An image histogram is a graphical representation of the distribution of pixel intensity values in an image. For a grayscale image, it shows how many pixels have each intensity value, generally from 0 to 255.

The histogram provides information about the brightness and contrast of an image. It can help identify whether an image is dark, bright, or has a narrow intensity range. A narrow distribution of intensity values may indicate low contrast

Q3. Differentiate between Histogram Stretching and Histogram Equalization.

Answer:

Histogram Stretching expands the existing pixel intensity range of an image to improve its dynamic range. It generally maps the minimum and maximum intensity values to a wider range.

Histogram Equalization redistributes pixel intensity values based on their frequency distribution to improve overall contrast.

In simple terms, histogram stretching expands the range of existing intensities, whereas histogram equalization redistributes intensities to produce a more balanced contrast.

Q4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

Answer:

CLAHE stands for Contrast Limited Adaptive Histogram Equalization. It enhances contrast by processing an image in smaller local regions instead of treating the entire image as one region.

Standard Histogram Equalization performs a global contrast enhancement on the complete image. CLAHE performs local enhancement and limits excessive contrast amplification.

Therefore, CLAHE is more suitable when different parts of an image have different illumination conditions

Q5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Answer:

Histogram equalization improves the contrast of an image and makes important visual details easier to distinguish. This can help reveal features that may be difficult to detect in a low-contrast image.

Better contrast can therefore provide clearer input for feature extraction and image segmentation, making important regions and boundaries easier to identify.

Q6. Mention three real-world applications where histogram equalization is widely used.

Answer:

Three real-world applications are:

Medical Imaging – improving the visibility of details in X-rays and other medical images. Satellite Imaging – enhancing land features and geographical details. Low-Light Photography – improving visibility in dark or poorly illuminated photographs.

These are examples of image-enhancement applications described in the course material.

Q7. What are the limitations of global histogram equalization?

Answer:

Global histogram equalization has some limitations:

It applies the same enhancement to the entire image. It may produce excessive contrast in some regions. It does not adapt well to images with uneven illumination. It can make existing noise or unwanted variations more noticeable.

Therefore, global histogram equalization may not always produce natural-looking results, especially when different regions of an image require different levels of enhancement.

Q8. How does contrast enhancement improve the performance of object detection and recognition systems?

Answer:

Contrast enhancement improves the visibility of important image features such as edges, shapes, and textures. When these features become clearer, computer vision algorithms can analyze the image more effectively.

Improved image clarity can therefore help object detection and recognition systems identify important objects and boundaries more reliably. Histogram-based enhancement is used to improve image visibility before further computer vision processing.

Q9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Answer:

Brightness adjustment mainly changes the overall brightness of an image by increasing or decreasing pixel intensity values.

Histogram-based enhancement techniques analyze the distribution of pixel intensities and modify that distribution to improve contrast.

Therefore, brightness adjustment mainly changes how bright or dark an image appears, while histogram-based techniques focus on improving the distribution of intensities and making image details more visible

Q10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

Answer:

CLAHE is useful for medical imaging and low-light images because it improves local contrast rather than applying the same enhancement to the entire image.

This is helpful when different parts of an image have different illumination levels. CLAHE can enhance local details while limiting excessive contrast amplification, making important structures easier to observe.

It is therefore useful for images where preserving local details and handling uneven illumination are important.

Double-click (or enter) to edit